Munchausen numbers

What is known about numbers equal to the sum
of their digits raised to themselves

$$3435 = 3^3 + 4^4 + 3^3 + 5^5$$

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An exposition of the library in `docs/library`,
written with Claude Code

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Preface

To be written.

Chapter 1

Definitions and first facts

1.1 The problem

To be written.

1.2 How long a solution can be

Proposition 1.1 (Length bound). PROVEN L1.2
A solution in base b has at most b digits.

Proposition 1.2 (Digit bound). PROVEN L1.2
A k-digit solution in base b uses only digits x with $x^x < b^k$.

To be written.

1.3 One base per digit string

Proposition 1.3 (One base per string). PROVEN L1.4
A digit string with at least two digits is a solution in at most one base.

To be written.

1.4 The two conventions for 0^0

Proposition 1.4 (The two conventions). PROVEN L1.5
$F_1(n) = F_0(n) + z(n)$, where $z(n)$ is the number of zero digits of n.

To be written.

1.5 Balance

To be written.

1.6 The digit-sum identity and other congruences

Proposition 1.5 (Digit-sum identity). PROVEN L1.6
Every solution satisfies $\sum_x (w(x) - x) \equiv 0 \pmod{b-1}$, the sum over its digits.

Proposition 1.6 (The only invariant residue). PROVEN L10.4
If $n \bmod M$ depends only on the digit multiset of n for all n with at least two digits, then M divides $b-1$.

Theorem 1.7 (Parity invariant). PROVEN L15.1
Let $b \geq 3$ and $q \geq 2$. Then $F(n) \equiv n \pmod{q}$ for every $n \geq 1$ if and only if $q = 2$, b is odd and $0^0 = 0$.

To be written.

1.7 Leading digits and solutions with b digits

To be written.

1.8 The data at a glance

To be written.

1.9 Earlier work

To be written.

Chapter 2

Constructions and families

2.1 Twins

Proposition 2.1 (Twins). PROVEN L2.1
Under $0^0 = 0$, a number n whose last digit is 0 is a solution if and only if $n + 1$ is a solution.

To be written.

2.2 Self-paying digits

Proposition 2.2 (Which bases have self-paying digits). PROVEN L2.2
Write $b = r^t$ with r not a perfect power. The self-paying digits of base b are $a = r^s$ with $1 \leq s < t$ and $t \mid s(r^s - 1)$, at position $k = s(r^s - 1)/t$.

Proposition 2.3 (Structure under \$0^0 = 0\$). PROVEN L2.2
Under $0^0 = 0$, every solution splits uniquely into a core, a set of self-paying digits on zero positions of the core, and an optional final 1. Every such combination is a solution.

To be written.

2.3 Swaps and chains

To be written.

2.4 Two-digit solutions

Proposition 2.4 (Two-digit solutions). PROVEN L2.4
The string (a, c) is a solution if and only if a divides $w(c) - c$ and $b = (a^a + w(c) - c)/a$ exceeds a and c. The base is then b.

Theorem 2.5 (Two 2-digit solutions). PROVEN L15.2
Under $0^0 = 1$, base 3 is the only base with two 2-digit solutions.

To be written.

2.5 Every length under $0^0 = 0$

Proposition 2.6 (Every length under \$0^0 = 0\$). PROVEN L2.5
Under $0^0 = 0$, every length $k \geq 2$ has infinitely many bases with a k-digit solution.

To be written.

2.6 Binomial families: every length under $0^0 = 1$

Theorem 2.7 (Binomial families). PROVEN L22.3
Let $j \geq 2$, $c \geq 1$, $e \geq 1$, $t_i = c^{j-i}\binom{j}{i}e^i$ for $0 \leq i \leq j-1$, and $x = e^j + \sum_i t_i^{t_i}$. If $x^x = y^j$ for an integer y, $c \mid y - e$ and $b = (y-e)/c$ exceeds every digit, then $(t_0, ..., t_{j-1}, x)$ is a solution in base b under both conventions.

Corollary 2.8 (Every length under \$0^0 = 1\$). PROVEN L22.3
For every $k \geq 3$ there are infinitely many k-digit solutions under $0^0 = 1$, and under $0^0 = 0$.

To be written.

2.7 Self-paying blocks

To be written.

2.8 Many solutions in one base

Proposition 2.9 (Unbounded counts under \$0^0 = 0\$). PROVEN L4.1 L9.8
For every $n \geq 3$, base n^{n-1} has at least $2^{n-1} - 2$ nontrivial solutions under $0^0 = 0$. So $N_0(b)$ is unbounded.

To be written.

2.9 Integers with several bases

To be written.

Chapter 3

The probabilistic model

3.1 The counts per base

To be written.

3.2 The basic model and the large-base limit

Heuristic 3.1 (The length law). HEURISTIC L5.2

In the large-base limit a base has on average one solution per convention before the parity effect. Its length is b, $b-1$, $b-2$, $b-3$ with probabilities about 8.0%, 55.2%, 23.2%, 8.6%, and shorter lengths are rarer by a factor of about e per digit.

To be written.

3.3 The digit-sum identity and the parity effect

To be written.

3.4 Testing the model by changing the weights

To be written.

3.5 Near-solutions and the defect spectrum

To be written.

3.6 A Poisson limit theorem

Theorem 3.2 (Poisson approximation). PROVEN L13.2

In the coupled model, the joint law of (N_0, N_1) is within total variation distance $O(1/b)$ of a bivariate Poisson law. The refined version needs an assumption on the residues of x^x.

To be written.

3.7 Empty bases

To be written.

3.8 Length against base

Conjecture 3.3 (Length window). CONJECTURE L10.5

Under $0^0 = 1$ and for every $\varepsilon > 0$, only finitely many solutions with $k \geq 3$ digits outside algebraic families lie in a base $b > k + \ln k + (1+\varepsilon)\ln\ln k$, and infinitely many lie in a base $b > k + \ln k + (1-\varepsilon)\ln\ln k$.

To be written.

3.9 Cycles of $n \mapsto F(n)$

To be written.

3.10 Anomalies and how they were resolved

To be written.

Chapter 4

Searching: algorithms and their limits

4.1 The prefix-pruned tree

To be written.

4.2 The cost of the exact search

To be written.

4.3 Inexact search with a measured miss rate

To be written.

4.4 An entropy bound for tree searches

Theorem 4.1 (Entropy bound). PROVEN L10.2

In the model, every search that walks the digit tree from the largest digit down and finds a solution with probability $1/2$ visits about 1.638^b nodes in large bases.

To be written.

4.5 Ideas that do not help

To be written.

4.6 Complexity

Proposition 4.2 (The real problem is in NP). PROVEN L8.1

The set of bases b, written in unary, that have a nontrivial solution is in NP.

Theorem 4.3 (Arbitrary weights). PROVEN L13.1

With arbitrary weights in place of x^x, deciding whether a solution exists is NP-complete, and counting the solutions is #P-complete.

To be written.

4.7 Short solutions are cheap

To be written.

4.8 Base 100

To be written.

Chapter 5

Short solutions and the Munchausen polynomial

5.1 The complete lists

To be written.

5.2 Local obstructions

Proposition 5.1 (Local obstruction). PROVEN L11.3
If the polynomial Q_s of a digit string s has no root modulo some integer $q \geq 2$, then s is not a solution in any base.

To be written.

5.3 Coverings and the residues of $x^x - x$

Theorem 5.2 (Covering by 27). PROVEN L15.4
Let $A(r) = \sum_i s_i r^{k-i}$ and $F_0 = \sum_i w(s_i)$ for a prefix $(s_1, \ldots, s_{k-1})$. If $A(r) - F_0 \not\equiv 0 \pmod 3$ and $A(r) - F_0 \not\equiv 2 \pmod{27}$ for every residue r modulo 27, then no string $(s_1, \ldots, s_{k-1}, x)$ with $x \geq 1$ is a solution in any base, under either convention.

To be written.

5.4 Galois groups and reducibility

To be written.

5.5 The root heuristic

To be written.

5.6 The square class and near-integers

Theorem 5.3 (Large leading digit). PROVEN L8.2
Fix $k \geq 2$ and $k-1$ lower digits below a bound B, and let the leading digit x vary with x^{x-1} a perfect $(k-1)$-th power. For x above an explicit bound, the only solutions are the self-paying families $(x, 0, \ldots, 0, 0)$ and $(x, 0, \ldots, 0, 1)$ under $0^0 = 0$, and $(x, 1)$ under $0^0 = 1$.

To be written.

5.7 Almost all bases have no short solution

Theorem 5.4 (Almost all bases). PROVEN L20.1
For every $\varepsilon > 0$, almost all bases b have no solution with $2 \le k \le (1-\varepsilon)\ln b / \ln\ln b$ digits, under either convention.

To be written.

5.8 Other roots: negative bases and rational roots

To be written.

5.9 Bases with a 2-digit and a 3-digit solution

To be written.

Chapter 6

The equation $b(b+x) = x^n + c$

6.1 Where the equation comes from

To be written.

6.2 Aurifeuillian factors and base 904

Theorem 6.1 (The ratio of Aurifeuillian factors). PROVEN L16.1

For squarefree odd $x > 1$, the Aurifeuillian factors $L < M$ of $x^x \pm 1$ satisfy $\ln(M/L) = 2\sum \psi(j)/\left(jx^{(j-1)/2}\right)$, the sum over odd $j \geq 1$ prime to x, with ψ the quadratic character of $\mathbb{Q}(\sqrt{x})$. In particular $M/L \to e^2$.

To be written.

6.3 Elementary exclusions and the limits of congruences

To be written.

6.4 The quadratic field

To be written.

6.5 Linear forms in logarithms

Theorem 6.2 (Prime powers). PROVEN L18.1

Let x be a prime power. The solutions of $b(b+x) = x^n + 1$ with $n \geq 2$ are exactly $(x, n) = (3, 2), (8, 2), (3, 3), (5, 3), (8, 3), (64, 3), (7, 7)$.

Corollary 6.3 (The diagonal \$n = x\$). PROVEN L18.1

If $b(b+x) = x^x + 1$ has a solution with x other than 3 and 7, then $x > 4 \cdot 10^9$ is odd, composite, not a square and not a prime power, and the class group of $\mathbb{Q}\left(\sqrt{x^2+4}\right)$ has an element of order $d \mid x$ with $x/d < 3.3 \cdot 10^4 \ln x$.

To be written.

6.6 Other exponents and the abc conjecture

To be written.

6.7 The even case

To be written.

6.8 The families $(1, x, 2)$ and $(1, x, 3)$

To be written.

6.9 Cores in perfect-power bases

To be written.

Chapter 7

Other number systems and other weights

7.1 A common frame

To be written.

7.2 Negative bases

To be written.

7.3 Imaginary, Gaussian and Eisenstein bases

To be written.

7.4 Rational bases

To be written.

7.5 A Pisot base

To be written.

7.6 Other weights

To be written.

7.7 Function fields, and why the model misses there

To be written.

7.8 The factorial number system

To be written.

7.9 What is universal

To be written.

Chapter 8

Open problems

8.1 Status

To be written.

8.2 Problems

To be written.

Appendix A

Probability: Poisson approximation, Borel-Cantelli and entropy

To be written.

Appendix B

Galois groups, Chebotarev and Jordan

To be written.

Appendix C

p-adic numbers and Hensel lifting

To be written.

Appendix D

Quadratic fields, Pell equations and cyclotomic factors

To be written.

Appendix E

Linear forms in logarithms

To be written.

Appendix F

The abc conjecture

To be written.

Appendix G

Elliptic and genus 2 curves

To be written.

Appendix H

Complexity: NP, #P and sparse sets

To be written.

Appendix I

Numeration systems

To be written.

Appendix J

Function fields over $\mathbb{F}_q$

To be written.

Appendix K

Tables of solutions

To be written.

Appendix L

Software and data

To be written.

Appendix M

Formalization in Lean

To be written.

Appendix N

Index of results

To be written.

Appendix O

Notation and glossary

To be written.

Bibliography